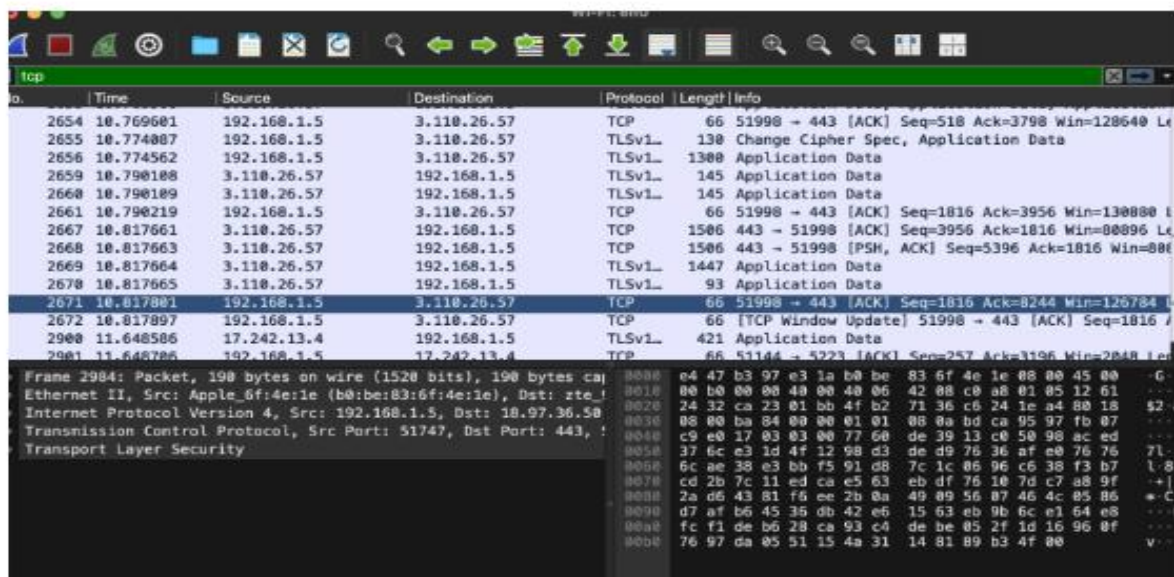


ASSIGNMENT:1

Introduction to Networking and the OSI Model Theory:

Task (1):



Task (2):

Packet	Protocol (Wireshark)	OSI Layer	Explanation
Packet 1	HTTP	Layer 7 – Application	HTTP is used by applications such as web browsers to request and receive web pages.
Packet 2	TCP	Layer 4 – Transport	TCP provides reliable, ordered delivery of data between applications.
Packet 3	IPv4	Layer 3 – Network	IP handles logical addressing and routing packets between networks.

Task (3):

OSI Layer	Layer Name	Daily-Life Example
7	Application	WhatsApp – messaging and file sharing
6	Presentation	Instagram – image/video formatting and encoding
5	Session	Google Meet – establishes and manages communication sessions
4	Transport	YouTube – TCP/UDP helps transport video data
3	Network	Google Maps – IP addressing and routing
2	Data Link	Wi-Fi/Ethernet – transfers frames between nearby devices
1	Physical	Ethernet cable/Wi-Fi radio – transmits electrical or radio signals

Task (4):

How a WhatsApp Message Travels Through the OSI Model:

OSI Layer	What Happens
7 – Application	You type “ Hello ” in WhatsApp and press Send. WhatsApp prepares the message to be sent over the network.
6 – Presentation	The message data is formatted and protected through encryption , helping keep the communication private.
4 – Transport	The message is broken into manageable pieces and transported using network transport protocols such as TCP where applicable, helping ensure data reaches the destination correctly.
3 – Network	IP addressing and routing are used to determine how the data should travel across the Internet toward the recipient's device.
2 – Data Link	Your phone puts the network data into frames and sends them through the local Wi-Fi or mobile network .
1 – Physical	The actual bits travel as radio signals over Wi-Fi/mobile networks or as electrical/optical signals through network infrastructure.